Prospective Randomized Controlled Trial Comparing Anesthetic Management With Remimazolam Besylate and Flumazenil Versus Propofol During Awake Craniotomy Following an Asleep-awake-asleep Method

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Background: Awake craniotomy is performed to resect brain tumors in eloquent brain areas to maximize tumor reduction and minimize neurological damage. Evidence suggests that intraoperative anesthetic management of awake craniotomy with remimazolam is safe. We compared the time to arousal and efficacy of anesthetic management with remimazolam and propofol during awake craniotomy.

Methods: In a single-institution randomized, prospective study, patients who underwent elective awake craniotomy were randomized to receive remimazolam and reversal with flumazenil (group R) or propofol (group P). The primary end point was time to awaken. Secondary end points were time to loss of consciousness during induction of anesthesia, the frequency of intraoperative complications (pain, hypertension, seizures, nausea, vomiting, and delayed arousal), and postoperative nausea and vomiting. Intraoperative task performance was assessed using a numerical rating scale (NRS) score.

Results: Fifty-eight patients were recruited, of which 52 (26 in each group) were available for the efficacy analysis. Patients in group R had faster mean ($\pm$SD) arousal times than those in the P group (890.8 $\pm$ 239.8 vs. 1075.4 $\pm$ 317.5 s; $P = 0.013$)and higher and more reliable intraoperative task performance (NRS score 8.81 $\pm$ 1.50 vs. 7.69 $\pm$ 2.36; $P = 0.043$). There were no significant intraoperative complications.

Conclusions: Compared with propofol, remimazolam was associated with more rapid loss of consciousness and, after administration of flumazenil, with faster arousal times and improved intraoperative task performance.

Key Words: awake craniotomy, flumazenil, propofol, prospective trial, remimazolam

(*J Neurosurg Anesthesiol* 2025;37:40–46)

Awake craniotomy (AC) has been reported to have beneficial outcomes for the resection of lesions in eloquent areas of the brain compared with similar surgeries performed with general anesthesia.[1,2] A clear arousal state is crucial for patients to perform tasks to guide surgical resection during the awake phase of AC.[3] Currently, propofol is widely used for anesthesia management during asleep-awake-asleep AC procedures.[3,4] Dexmedetomidine, a highly selective α2 agonist, is often used during monitored anesthesia care for AC because of its sedative and analgesic effects, and ability to maintain spontaneous breathing.[4]

Remimazolam besylate, a new ultra-short-acting benzodiazepine, can be used as a sedative or anesthetic agent and was introduced into clinical practice in Japan in 2020. Remimazolam allows for the safe management of general anesthesia in patients with various diseases, along with propofol, midazolam, and inhalation agents.[5] Being a benzodiazepine, the effects of remimazolam can be reversed by flumazenil. It has been suggested that the anesthesia management of AC can be safely performed using remimazolam[6] and remimazolam reversed by flumazenil.[7] In a previous retrospective study, we did not find a significant difference in the time to awakening between patients who received remimazolam or propofol for AC.[7] However, in that study, flumazenil was administered to reverse remimazolam only in patients who were not fully awake upon recovery of spontaneous breathing. No prospective studies have compared the efficacy of remimazolam (and reversal with flumazenil) with propofol in the anesthesia management of AC. In the current study,

Received for publication January 31, 2024; accepted April 23, 2024.
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T.S. and K.M.: substantially contributed to the study design. T.S.: collected and analyzed the data and wrote the manuscript. T.S., T.A., I.A., and K.O.: performed anesthesia. T.S., Y.K., and M.A.: performed the statistical analyses. T.S.: wrote the manuscript.
Support for the study was provided solely from institutional and/or departmental sources.
The authors have no conflicts of interest to declare.
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DOI: 10.1097/ANA.0000000000000975

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we compared the use of remimazolam and propofol during AC and investigated whether remimazolam with flumazenil reversal allows earlier awakening than propofol.

METHODS

This single-institution, randomized, and prospective trial was approved by Nagoya University (2020-0494; January 13, 2021) and registered in the Japan Registry of Clinical Trials (jRCT1041200094; January 28, 2021). Written informed consent was obtained from all participants. The study was conducted in accordance with the Declaration of Helsinki.

Adult patients who underwent elective AC at Nagoya University Hospital between February 2021 and March 2023 were enrolled in this study. Inclusion criteria were (1) brain tumor in the frontal or temporal lobe requiring intraoperative language tasks or focal resection for epilepsy near the language areas, requiring the patient to be awake during the resection; (2) age between 18 and 70 years; and (3) written consent to participate in the study. Exclusion criteria were (1) American Society of Anesthesiologists physical status score above 3; (2) liver failure (Child-Pugh classification B or C); (3) severe respiratory disease (eg, severe obstructive pulmonary disease); (4) hemostatic dysfunction (platelet count $<80{,}000$, PT-INR >1.5, or APTT >40 s); (5) serious respiratory dysfunction (vital capacity $<60\%$ or forced expiratory volume in 1 s $<50\%$); 6) angle closure glaucoma; (7) decreased cardiac function (ejection fraction $<45\%$, presence of severe valvular disease, or a history of coronary artery disease); and (8) refusal to participate.

Randomization and Blinding

The data center of the Department of Advanced Medicine, Nagoya University Hospital, randomized the patients in a 1:1 ratio to receive either remimazolam (group R) or propofol (group P). Group allocation was performed using an online system managed by the data center, which also issued IDs and passwords to control access. Randomization was conducted using the minimization method with age (below 40 and 40 y or above), sex (male and female), height (<165 and ≥ 165 cm), and weight (<60 and ≥ 60 kg) as minimization factors.

Anesthesia Management

All patients underwent AC using an asleep-awake-asleep protocol. In group R, general anesthesia was induced with a 12 mg $kg^{-1}{\cdot}h^{-1}$ continuous infusion of remimazolam besylate (Anerem, Mundipharma K.K.), which was reduced after loss of consciousness. In group P, anesthesia was induced with propofol (1% Diprivan Injection-kit, Aspenpharma, Tokyo, Japan) administered as a 3-μg/mL target-controlled infusion. Fentanyl (25 to 100 μg) and remifentanil (0.1 to 0.2 $\mu g{\cdot}k^{-1}{\cdot}min^{-1}$) were administered in both groups. Anesthesia was maintained with remimazolam (0.5 to 1.0 $mg{\cdot}kg^{-1}{\cdot}h^{-1}$) or propofol (2.5 to 4.5 $\mu g{\cdot}mL^{-1}$ of target control infusion) according to group allocation, in conjunction with remifentanil (0.1 to 0.25 $\mu g{\cdot}kg^{-1}{\cdot}min^{-1}$). Small doses of rocuronium (10 to 30 mg) were administered during induction and maintenance of anesthesia as required. An I-gel (size: male #4, female #3) supraglottic airway device (Intersurgical) and pressure-regulated volume control ventilation was used in all patients during the asleep phase of the AC.

Anesthesia was adjusted to maintain the bispectral index (BIS monitor, Philips, Japan) between 40 and 60. Remifentanil was titrated to maintain systolic blood pressure below 140 mm Hg during the asleep phase. SpO_2 was maintained $\geq 94\%$ by adjustment of the inspired oxygen fraction throughout. Heart rate was maintained between 50 and 100 bpm by adjustment of depth of anesthesia and use of beta-blockers for tachycardia or atropine sulfate for bradycardia as appropriate. Bilateral scalp blocks (supraorbital, supratrochlear, greater and lesser occipital, auriculotemporal, and zygomaticotemporal nerves) were performed to minimize intraoperative pain.[8]

Levetiracetam (500 mg) was administered as an antiepileptic agent,[9] and a low dose of dexamethasone (6.6 mg) as an antiemetic[10] before the awake phase of the procedure. Twenty percent mannitol (150 to 300 mL) was administered to all patients.

The anesthetic infusions were terminated when the dura was exposed. The supraglottic airway was removed after confirmation of the following signs of arousal as previously described:[7] (1) recovery of spontaneous breathing, defined as a respiratory rate >8 breaths/min, tidal volume >5 mL kg^{-1} or >300 mL; (2) the patient was not agitated; (3) the patient was able to respond to verbal commands, such as instructions to grip the investigator's hand, flex a foot, or eye open; and (4) BIS >75. Patients in the R group were administered flumazenil after recovery of spontaneous breathing, and 2 of the above arousal symptoms were confirmed.

Following awakening, metoclopramide (10 mg) was administered if the patient complained of nausea. Local anesthetic (1% lidocaine with epinephrine 1:100,000) infiltration of the scalp wound and flurbiprofen (50 mg) or a low dose of fentanyl (25 to 50 μg) were administered for rescue analgesia as required.

Following tumor resection, all patients were resedated with remimazolam or propofol according to group allocation. Patients were transferred to the intensive care unit after the operation.

Data Collection

BIS values during the initial asleep phase of the AC were recorded by averaging the values 4 times: after insertion of the supraglottic airway device, at the time of patient positioning, at the start of surgery, and at the time of dural incision.

Complications during the awake phase (pain, hyper/hypotension, tachy/bradycardia, vomiting, seizures, and agitation) and nausea and vomiting were recorded. Patients who complained of headaches during the awake phase were classified as having intraoperative pain (self-assessed numerical rating scale [NRS] score >4).[10] Patients were classified as hypertensive if blood pressure was $>20\%$ above baseline value or systolic blood pressure was

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>150 mm Hg, and hypotensive if blood pressure was <20% baseline or systolic blood pressure <80 mm Hg.[11] Agitation was defined as a Richmond Agitation-Sedation Scale score >2.

Delayed arousal was defined as no signs of arousal 30 minutes after the anesthetic infusion was stopped. Intraoperative task performance was evaluated by a speech therapist unaware of group allocation. The ability of the patient to cooperate and perform intraoperative mapping was assessed using a 10-point NRS score (0 = unsatisfactory; 10 = excellent). Intraoperative cortical mapping was considered unsuccessful if the NRS score was <6. Postoperative nausea and vomiting were defined as the occurrence of nausea or vomiting within 24 hours postsurgery.

Outcomes

The primary outcome was the time to awakening, defined as the time from the discontinuation of anesthesia to the removal of the supraglottic airway. Secondary outcomes were the frequency of intraoperative complications (pain, hypertension, seizures, nausea, vomiting, agitation, and delayed arousal), intraoperative task performance, and postoperative nausea and vomiting occurrence.

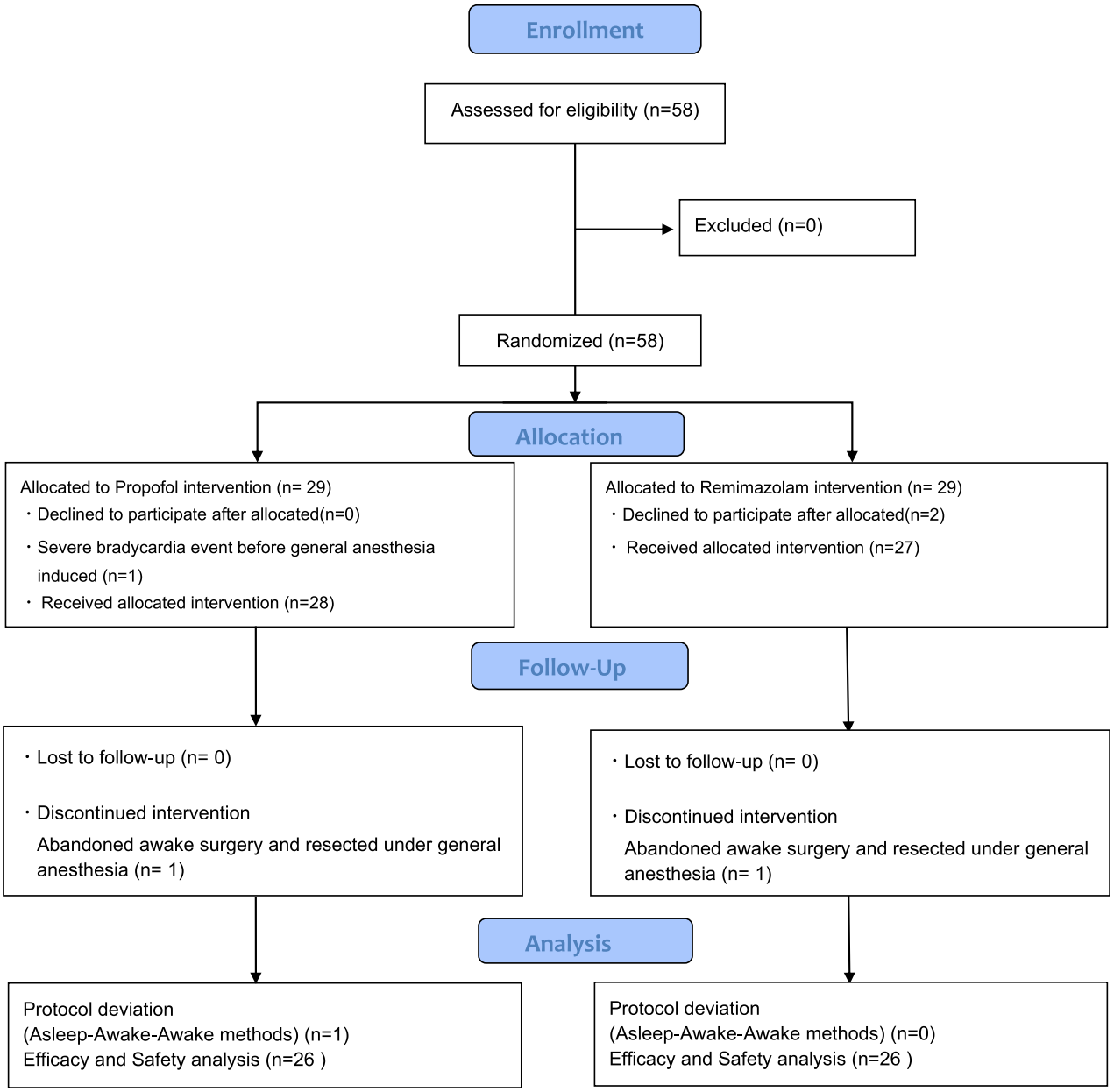


FIGURE 1. Study flow diagram. full color online

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Sample Size and Statistical Analysis

Analyses of the primary outcome were performed using STATA, R, and EZR software.[12] The primary outcome was compared between the two groups using analysis of covariance with age, gender, height, and weight as covariates, as well as secondary outcomes of hemodynamic changes during supraglottic airway insertion (heart rate and blood pressure values and their rate of change immediately before insertion and 1 min after insertion), intraoperative pain score (NRS), amount of fentanyl use, and intraoperative task performance.

The secondary outcome, complications during the awake phase, was assessed using chi-square tests and the Mann-Whitney *U* test, as appropriate. The mean BIS value was analyzed by *t* test. All data are presented as mean ± SD or median (interquartile range). $P < 0.05$ was considered statistically significant.

Our previous analysis found an arousal time prior to the awake phase of AC with propofol of 1138 ± 374 seconds (~19 min).[7] Considering a 25% reduction in arousal time (300 s) to represent a clinically significant difference and assuming the difference in time between the 2 groups to be 80% of the SD, the required sample size was 26 patients per group to detect a significant difference with $\alpha = 0.05$ and a power of 80%. Considering potential discontinuation and withdrawal, we aimed to include 29 patients per group; thus, it was planned to include 58 patients in total.

RESULTS

Fifty-eight patients were recruited, of which 2 patients withdrew their consent to participate after randomization. A further patient was excluded due to severe bradycardia before anesthesia, and 2 patients were excluded because the intervention was discontinued secondary to bleeding or severe brain swelling. Of the remaining 53 patients, 1 patient in group P was subsequently excluded because of a protocol deviation; AC was performed using an asleep-awake-awake technique. Therefore, 52 patients (26 in each group) were available for the efficacy analysis (Fig. 1).

There were no significant differences in patient demographics or tumor location between the 2 groups (Table 1). All patients successfully completed AC, and all patients in the R group were administered flumazenil as an antagonist.

TABLE 1. Patient Data

	Group P (n = 26)	*Group R (n = 26)*	*P*-value
Age (y)	41.5 [34.0–51.0]	41.0 [32.0–50.0]	0.79
Sex (male/female)	16/10	17/9	0.77
Height (cm)	167.4 [156.2–175.7]	170.3 [163.0–174.4]	0.38
Weight (kg)	59.4 [52.8–70.4]	64.0 [56.2–75.3]	0.16
BMI (kg/m^2)	21.6 [19.6, 24.1]	22.50 [20.3, 24.5]	0.29
Lesion of tumor (anterior /temporal)	19/7	20/6	0.75

Data presented as median [interquartile ranges] or number of patients.
Group P indicates propofol; Group R, remimazolam.

Remimazolam induced loss of consciousness more quickly during induction of anesthesia compared with propofol (105.5 ± 19.82 [95% CI, 97.5-113.4] s vs. 252.1 ± 99.32 [95% CI: 211.9-292.2] s; $P < 0.001$). Remimazolam was also associated with faster arousal times than propofol (890.8 ± 239.8 [95% CI: 793.9-987.6] s vs. 1075.4 ± 317.5 [95% CI, 947.2-1203.6] s; $P = 0.013$) (Fig. 2). Patients in group R tended to perform significantly better during intraoperative task performance and language assessment (NRS 8.81 ± 1.50 vs. 7.69 ± 2.36; $P = 0.043$) (Fig. 3). Mean BIS was similar between the 2 groups (50.7 ± 6.8 [95% CI: 47.9 to 53.4] vs. 48.3 ± 5.8 [95% CI, 45.9-50.6] in group R and group P, respectively; $P = 0.17$). There were no significant intraoperative complications in either group (Table 2), and no patient had delayed arousal.

DISCUSSION

AC is indicated in patients for resection of lesions in eloquent brain regions, including language areas, to minimize language dysfunction and other neurological deficits and improve patient outcomes compared with resection with general anesthesia.[1,13,14] Our study found that a combination of remimazolam and flumazenil was associated with faster arousal times during AC compared with propofol. In addition, remimazolam was associated with clearer arousal, allowing better performance during intraoperative language tasks.

For safe intraoperative management, the patient should be fully conscious and cooperative during the awake phase of the procedure. Several anesthetic techniques have been described to facilitate this. Propofol is widely used in combination with remifentanil and has ideal pharmacokinetics for intraoperative AC management.[15] Propofol is a gamma-aminobutyric acid receptor agonist with many pharmacological advantages over other anesthetic agents, including rapid onset of action, short duration of action, and fewer side effects such as postoperative nausea.[15,16] However, propofol infusions have side effects, including pain on injection, bradycardia, hypotension, and hyperlipidemia. The fact that propofol has no antagonist is potentially problematic when rapid and clear awakening is required. Some patients cannot cooperate during AC because of delayed arousal or inability to accomplish language tasks. Dexmedetomidine, a highly selective α2-adrenergic receptor agonist with sedative, analgesic, anxiolytic, and sympatholytic effects, is often used during monitored anesthesia care for AC.[17,18] In contrast with other sedatives, dexmedetomidine is not associated with respiratory depression.[4,18] However, dexmedetomidine may cause deep sedation, which can make verbal tasks difficult, and, like propofol, has no antagonist.

Remimazolam is a novel ultra-short-acting benzodiazepine agent with rapid onset of action, short duration of action, and rapid recovery compared with other available benzodiazepines, such as midazolam and diazepam.[5,19,20] It has an onset time of approximately 1 to 3 minutes and a short metabolic half-life, thereby pro-

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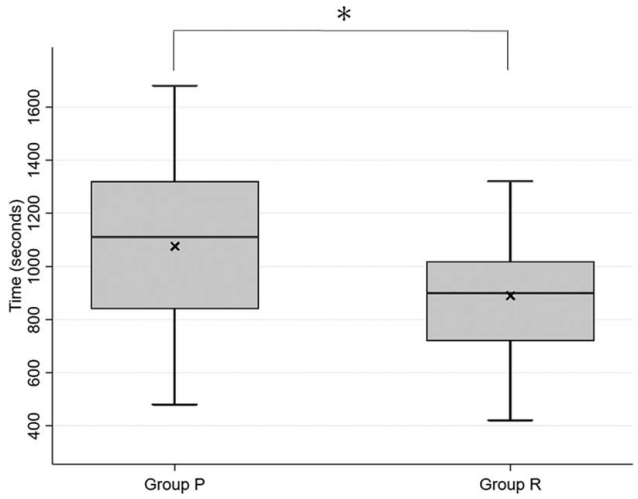


FIGURE 2. Box and whisker plots showing intraoperative awakening times. The boxes represent the median (solid line) and 25th and 75th percentiles, and the whiskers represent the upper and lower extremities. X shows the mean value. *$P < 0.05$ in the analysis of covariance. Group P, propofol; Group R, remimazolam/flumazenil.

viding moderate anesthesia but fast recovery.[19] Remimazolam also has the advantage of being antagonized by flumazenil. The half-life of remimazolam is approximately the same as that of flumazenil, allowing the safe reversal of the anesthetic effects of remimazolam with flumazenil.[19]

We previously assessed the effectiveness of remimazolam in AC in a retrospective study,[7] but the current study is the first prospective randomized controlled trial to investigate remimazolam use in AC. In this study, patients who received remimazolam (reversed with flumazenil) awoke significantly more quickly than those who received propofol. A previous study also reported faster recovery times in patients undergoing rigid bronchoscopy with remimazolam general anesthesia (reversed with flumazenil) compared with propofol and no differences in hemodynamic changes or adverse events between the 2 agents.[21] These findings suggest that remimazolam and its reversal with flumazenil allows faster and more reliable awakening than propofol anesthesia, making the remimazolam/flumazenil combination a potentially useful anesthetic technique for AC.

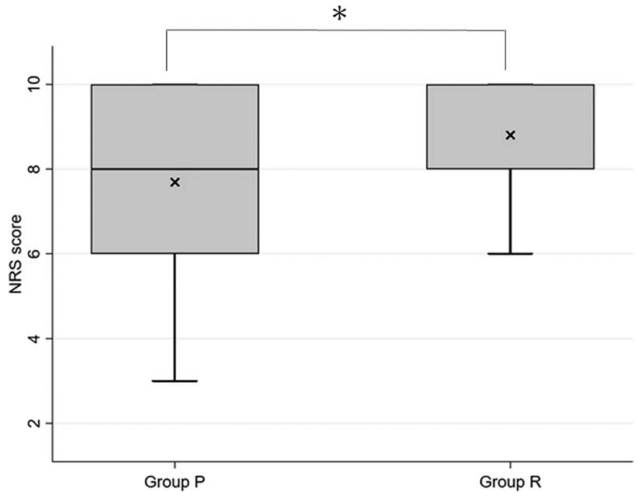


FIGURE 3. Box and whisker plots showing intraoperative performance task achievement. The boxes represent the median (solid line) and 25th and 75th percentiles, and the whiskers represent the upper and lower extremities. X shows the mean value. *$P < 0.05$ in the analysis of covariance. Group P, propofol; Group R, remimazolam/flumazenil. NRS, numerical rating scale of the ability of the patient to cooperate with intraoperative performance tasks.

Benzodiazepines, including midazolam, diazepam, and remimazolam, have sedative and anxiolytic effects when administered intravenously, and these effects are rapidly and effectively antagonized by flumazenil.[19,22] Occasional restlessness and agitation has been reported following reversal of remimazolam anesthesia with flumazenil,[19] and 1 case of resedation has been reported when remimazolam anesthesia was reversed with flumazenil.[23] However, no restlessness, agitation, or resedation events were observed in this study.

We based our criteria for removal of the supraglottic airway removal after awakening in this study on our previous report.[7] Flumazenil was administered only after spontaneous breathing and body movement had been confirmed after cessation of remimazolam. Using this approach, no resedation was observed in our study. Spontaneous respiration can be expected to be maintained in the event of resedation after the reversal of remimazolam with flumazenil, at least with the appropriate partial pressure of exhaled carbon dioxide, provided that spontaneous respiration was restored before reversal. However, there might be a risk of hypoxemia if the patient's arousal level is insufficient to allow spontaneous respiration before reversal of remimazolam with flumazenil. Although the likelihood of severe hypoxemia in the event of resedation is considered to be low, it is safer to wait until the patient has recovered some level of arousal following cessation of remimazolam anesthesia before administering flumazenil.

In this study, there was a trend for patients in the R group to have higher NRS scores for intraoperative task performance and language assessment by a speech therapist. A few patients in group P had a low NRS score of 3, and some exhibited poor arousal, whereas the lowest NRS score recorded in the R group was 6. We presume that this difference was related to the reversal of the anesthetic effect of remimazolam with flumazenil resulting in higher arousal levels. Remimazolam is a new drug and its cost is currently higher than that of existing drugs used during AC, such as propofol. However, as the combination of remimazolam and flumazenil can provide a more reliable and higher-quality awakening during AC, it could be considered cost-effective.

In general, nausea and vomiting during AC are problematic because of the potential for increased intra-

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TABLE 2. Intraoperative Data

	Group P (n = 26)	Group R (n = 26)	*P*
Operation time (min)	448.0 [399.0-522.0]	443.0 [384.0-518.0]	0.69
Anesthesia time (min)	550.0 [499.0-632.0]	542.5 [502.0-624.0]	0.74
Awake time (min)	116.0 [93.0-132.0]	124.0 [101.0-154.0]	0.31
sABP before SGA insertion (mm Hg)	101.0 ± 19.5	109.9 ± 16.1	0.09
sABP after SGA insertion (mm Hg)	106.1 ± 20.3	108.6 ± 19.5	0.87
dABP before SGA insertion (mm Hg)	51.6 ± 12.7	53.9 ± 12.5	0.47
dABP after SGA insertion (mm Hg)	52.5 ± 15.8	56.7 ± 13.8	0.36
HR before SGA insertion (beats.min^{-1})	73.5 ± 10.6	80.7 ± 15.3	0.07
HR after SGA insertion (beats.min^{-1})	75.4 ± 9.2	82.8 ± 17.9	0.09
Total infusion before awake phase (mL)	1550.0 [1400.0-1950.0]	1725.0 [1450.0-2100.0]	0.37
Bleeding (mL)	315.5 [141.0-512.0]	294.0 [140.0-586.0]	0.94
Dose of fentanyl before awake phase (μg)	75.0 [50.0-100.0]	50.0 [25.0-100.0]	0.036
HT	10 (38.4)	14 (53.8)	0.26
Nausea	5 (19.2)	7 (26.9)	0.51
Vomiting	1 (3.8)	2 (7.7)	0.55
Pain (NRS scale)	5.0 [2.5, 6.5]	5.0 [3.0, 6.0]	0.46
Seizure	10 (38.5)	8 (30.8)	0.56
LOC time (s)	228.0 [169.0-341.0]	98.5 [92.0-118.0]	< 0.001
LOC TCI concentration (μg.mL^{-1})	2.5 [2.1-2.9]	—	—
LOC dose of remimazolam (mg)	—	23.4 [20.2–26.0]	—
Dose of flumazenil (mg)	—	0.3 [0.3, 0.5]	—
Mean BIS value	48.3 ± 5.82	50.7 ± 6.80	0.17
24-h PONV	10 (38.5)	7 (26.9)	0.38

Data presented as median [interquartile range], mean ± SD, or number (%) of patients

dABP indicates diastolic arterial pressure; Group P, propofol; Group R, remimazolam; HR, heart rate; HT, hypertension; LOC, loss of consciousness; sABP, systolic arterial pressure; SGA, supraglottic airway.

cranial pressure, aspiration, and asphyxia. 5HT3 receptor inhibitors, such as ondansetron or granisetron, are routinely administered in many centers to minimize the risk of nausea and vomiting during the awake phase of AC.[2,3] However, 5HT3 receptor antagonists are not available for this indication in Japan and can only be used off-label. Therefore, we used metoclopramide and dexamethasone as alternative drugs for nausea prevention according to usual practice.[10]

This study has some limitations. First, only data from a single center were collected and analyzed; our ability to conduct a multicenter trial was hindered by the COVID-19 pandemic and a shortage of remimazolam in Japan. Second, this was not a double-blind study. Remimazolam is a clear, colorless liquid, whereas propofol is a white emulsion, making it difficult to mask the 2 drugs.[19,20] In addition, arousal times were assessed by the anesthesiologist who was aware of group allocation and assessors of other outcomes were also not blinded. Third, as of 2021 there were no countries other than Japan and South Korea where remimazolam was available for use as a general anesthetic. Bias may exist because the study was conducted in a single Japanese institution; however, it is anticipated that future research will be conducted collaboratively at multiple institutions. There is also concern that flumazenil could lower the seizure threshold and potentially increase the risk for epileptic brain activity and clinical seizures.[18,19] Though no convulsive seizures related to flumazenil administration were observed in this study, this risk remains. Caution should be exercised when administering flumazenil to reverse the anesthetic effect of remimazolam, particularly during AC for epilepsy. Though we aimed to ensure a similar depth of intraoperative sedation in both groups by adjusting anesthesia to maintain BIS between 40 and 60, remimazolam may produce higher values with BIS and Sedline monitoring,[24–26] potentially resulting in different depths of anesthesia between the 2, groups; patients in group R may have been more deeply sedated before the awake phase than those in P group. While the BIS monitor can objectively measure sedation level with the aid of EEG,[25] there is no other objective measurement. As such, we used BIS to assess the depth of anesthesia in this study. We did not measure intraoperative amnesia, but there were no complaints from patients of unpleasant memories during the AC procedure at postoperative rounds. Previous reports indicate that the time required to awaken during AC during an asleep-awake-awake protocol ranges from 10 to 25 min.[27] As this is almost identical to the time taken to awaken in this study, we conclude that patients in neither group were oversedated.

In conclusion, remimazolam and reversal with flumazenil were associated with faster awakening times compared with propofol during AC. The combination of remimazolam and flumazenil may also provide clearer arousal than propofol. This study suggests that remimazolam (and reversal with flumazenil) could be a useful anesthetic for an asleep-awake-asleep AC technique.

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